

UTDsolv Project Charter

General Information

The table below shows the high-level information for this project.

Client Information

Project Name	Converting Handwritten Jewelry Records into Digital Data
Project Budget	\$7,488
Project Deadline	08/28/2026 - 12/07/2026
Company	Lapidary Arts Jewelry
Company URL	https://www.lapidaryartsjewelry.com/
Sponsor Name and Role	Rania Eldisi - Project Sponsor
Phone	(214) 280-9733
Email	rania@lapidaryartsjewelry.com

UT Dallas – Jindal School of Management Team

Instructor Name	Senthil Velayudham
Project Manager	Adam Torres
Project Team Members	Adam Torres, Arnav Prabhu, Zainab Kazi, Varsha Janumpally, Hana El Mahdy, Anthony Alston
Date Prepared	09/08/2026

Version History

Ver. No	Ver. Date	Revised by	Description
V1.0	09/08/2026	UT Dallas Capstone Team	Initial charter based on proposal and 09/04/2026 sponsor meeting
V1.1	09/11/2026	UT Dallas Capstone Team	Added Price as sixth extraction field and documented Supabase, Vercel, and Transformers.js technical stack.
V1.2			

Project Background

Lapidary Arts Jewelry is a family-owned fine jewelry store in Plano, Texas that has served the Dallas-Fort Worth community since 1976. The company maintains historical handwritten and paper-based records, including completed invoices dating back to the 1990s. This project will develop a minimum viable digitization system that allows store staff to capture or upload historical records, extract key information using Transformers.js for OCR/AI-assisted handwriting recognition, review and correct the extracted data, store approved information in a structured Supabase database, host the web application on Vercel, and export records for future business use.

Client Business Needs/Purpose and Justification

Lapidary Arts has valuable historical customer and transaction information stored in handwritten paper records that are difficult to search, organize, analyze, or reuse. Digitizing these records will make historical information more accessible while reducing manual data-entry effort. The project will provide a practical workflow for converting completed paper invoices into structured digital records while preserving human verification for

accuracy and customer confidentiality. Verified data can then be searched, organized, and exported in CSV format for future use with external CRM or business systems.

Business Objectives

- Develop a working prototype for digitizing historical, completed paper invoices.
- Extract the six mandatory fields: Client Name, Phone Number, Date, Date Promised, Instructions/Article Details, and Price.
- Provide a human-in-the-loop interface for staff to review, edit, and confirm OCR-extracted information.
- Store verified records in a structured, searchable Supabase database, host the web application on Vercel, use Transformers.js for OCR/AI-assisted extraction, and enable CSV export for future external use.

Preliminary Project Scope (high-level)

The project is limited to a minimum viable system for digitizing historical, completed Lapidary Arts paper invoices. The team will focus on document capture/upload, OCR/AI-assisted extraction of six sponsor-defined fields using Transformers.js, human review and correction, structured online storage in Supabase, search/basic filtering, Vercel-hosted web access, and CSV export. The project does not include processing the company's entire archive, replacing the current POS system, managing active transactional orders, or directly integrating with live customer, payment, inventory, or CRM systems during the semester.

In Scope	Out of Scope
Capture/upload historical completed invoices	Process the entire company archive
OCR/AI extraction of six mandatory fields using Transformers.js	Manage active transactional orders
Human review/edit/confirmation	Replace the current POS system
Supabase storage with search/basic filtering	Direct integration with live customer, payment, inventory, or CRM systems
CSV export of verified records	Ongoing production support after capstone handoff

Key Milestones and Deliverables

The project will progress through requirements and architecture planning, prototype development, workflow implementation, testing and refinement, and final client handoff. Intermediate dates below are proposed planning dates within the confirmed project period and may be adjusted with the sponsor and instructor.

Milestone	Deliverable	Estimated End Date
Requirements & Architecture	Refined scope; six data fields; workflow, Supabase schema, and Vercel architecture design	09/18/2026
Core Prototype	Document/mobile capture; Transformers.js OCR ingestion; initial Supabase integration	10/16/2026
Review, Search & Export	Vercel-hosted verification/editing interface; search/filtering; CSV export	11/06/2026
Testing & Refinement	Six-field OCR accuracy testing; workflow validation; fixes and privacy review	11/20/2026

Milestone	Deliverable	Estimated End Date
Final Handoff	Working Vercel-hosted proof of concept; Supabase schema; sample results; workflow documentation; implementation recommendations	12/07/2026

Project Assumptions

The project team developed the following assumptions based on the current project scope, discussions with Lapidary Arts, and the requirements identified during the September 4 client meeting. These assumptions represent conditions the team expects will remain true throughout the project but may require further validation as development and testing progress.

- Lapidary Arts will provide the team with access to a sufficient number of approved historical paper invoices or sample records for development and testing.
- The sample documents provided will reasonably represent the different handwriting styles, layouts, and record conditions found within the historical invoice archive.
- The team will receive the necessary client feedback throughout the semester to validate extracted fields, system functionality, and usability.
- Any actual customer records used by the team will only be accessed and processed after the required NDA and data-handling requirements are completed.
- Transformers.js, Supabase, and Vercel will remain accessible and feasible within the project's technical, budget, and semester constraints.
- The six required data fields - Client Name, Phone Number, Date, Date Promised, Instructions/Article Details, and Price - will remain the primary extraction requirements unless the sponsor approves changes to the project scope.
- The project team will have sufficient access to the client and store workflow information needed to understand how historical invoices should be captured, reviewed, stored, and exported.

Project Critical Success Factors (CSF)

The project will be successful if the team delivers a functional prototype that enables Lapidary Arts to capture historical invoices, extract required information, verify and correct results, organize approved records digitally, and export the data while protecting customer confidentiality and providing clear documentation for future use.

- Functional end-to-end workflow from capture/upload through verified storage and CSV export.
- Accurate and reviewable extraction of the six sponsor-defined mandatory fields from approved samples.
- User-friendly human verification and correction workflow.
- Customer confidentiality and compliance with NDA/data-handling expectations.

Key Stakeholders

Key stakeholders were identified from the client, course leadership, and project team roles. The sponsor provides business requirements and feedback, the instructor provides academic oversight, the project manager coordinates execution and communication, and team members contribute to technical and business deliverables.

Role	Resource Name	Organization	Responsibilities
Project Sponsor	Rania Eldisi	Lapidary Arts Jewelry	Provide requirements, workflow context, feedback, and client approval.
Instructor	Senthil Velayudham	UT Dallas	Provide academic oversight, guidance, and escalation support.
Project Manager / Team Lead	Adam Torres	UT Dallas	Coordinate activities, meetings, responsibilities, documentation, and sponsor communication. Export functionality, End-to-end technical architecture, and database schema implementation.
Secondary PM / Team Member	Arnav Prabhu	UT Dallas	Support coordination and assigned project deliverables. Document upload/mobile capture integration, OCR/AI ingestion pipeline setup, and technical task tracking with the PM.
Team Member	Anthony Alston	UT Dallas	Designing extraction test cases, benchmarking OCR/AI accuracy across handwriting styles, and logging error patterns across the five target fields.
Team Member	Zainab Kazi	UT Dallas	Defining business validation rules, mapping CSV export schemas for future CRM/business system alignment, and coordinating data quality audits.
Team Member	Varsha Janumpally	UT Dallas	Mapping staff review and editing workflows, defining UI verification wireframes/acceptance criteria, and coordinating usability testing with the sponsor.
Team Member	Hana El Mahdy	UT Dallas	Managing NDA compliance protocols and customer data confidentiality standards, drafting staff workflow documentation, and designing training guides for final handoff.